CAPSTONE PROJECTS STATUS REPORT

AI-Ready Python Full Stack Engineering Program
Target: BE / B.Tech Artificial Intelligence & Machine Learning (AI/ML)

SUBMITTED IN FULFILLMENT OF THE CAPSTONE PHASE EVALUATION

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Program Duration:	15 Days / 90 Contact Hours (09:00 AM – 05:00 PM Daily)
Submission Date:	July 16, 2026

Executive Summary

This report outlines the technical achievements and evaluation status of the Capstone Projects executed during the final phase (Days 13–15) of the 15-day intensive AI-Ready Python Full Stack Engineering program. Conducted for the seventh-semester BE/B.Tech Artificial Intelligence & Machine Learning (AI/ML) students at Alvas Institute of Engineering & Technology, this phase required students to synthesize their learning of asynchronous APIs, stateful frontend components, vector databases, and containerized deployment.

To ensure high standards of academic rigor, originality, and practical competency, the students were divided into 17 teams and tasked with building unique, custom applications. Rather than relying on a pre-packaged template, each team designed, implemented, and deployed its own system. The final projects integrated secure JWT authentication, Neon cloud PostgreSQL databases, conversational LLM agents (LangChain), local embeddings generation, Qdrant vector databases, and multi-service Docker container orchestration deployed to public URLs on AWS, Vercel, and Render.

Every single team demonstrated a strong work ethic and contributed effectively, meeting the technical objectives of the program. On Day 15, the teams presented their functional, deployed capstone applications to their own college faculty panel, demonstrating solid technical understanding and readiness for engineering roles.

Capstone Project Guidelines & Requirements

To qualify for graduation, each team project was required to implement the following core features:

- Backend Architecture: FastAPI backend structured with modular routers, schemas, models, and Uvicorn server configs.
- Frontend Interface: Typed React 18 + TypeScript interfaces utilizing state hooks and unified Axios client service layers.
- Relational Persistence: PostgreSQL tables (users, files, activities) hosted on Neon DB, queried via SQLAlchemy async sessions.
- User Security: Password hashing using bcrypt, JWT token generation, and secure frontend/backend route guards.
- Generative AI Features: Memory-enabled chatbots (using LangChain and Groq) capable of conversational context retention.
- Semantic Search / RAG: File chunking, embedding generation using FastEmbed, and similarity vector queries in Qdrant Vector DB.
- Local Containerization: Standard Dockerfile definitions for both backend and frontend, orchestrated using Docker Compose.

- Production Deployment: Live application hosting utilizing AWS EC2/S3, Vercel, and Render environments.

Recommended Repository Structure

Teams maintained a structured layout to support clean container orchestration:
• backend/ - Routers, services, models, schemas, Dockerfile, and environment config.
• frontend/ - Source components, styling, pages, Axios services, and frontend Dockerfile.
• docker-compose.yml - Orchestrates backend, frontend, and local services.
• README.md - Documents installation steps, architecture diagrams, and live cloud URLs.

System Architecture & Evaluation Checklist

The typical data flow and infrastructure mapping of the capstone solutions is structured as follows:

```
                      Browser (React 18 + TS on Vercel)
                            │ (Axios HTTPS + Bearer JWT)
            FastAPI Backend (Docker Container on Render / AWS EC2)
                     ├── [Relational] ──> Neon Cloud PostgreSQL
      └── [Semantic/RAG] ─> FastEmbed ──> Qdrant Cloud Vector DB ──> Groq LLM
```

Evaluation Checklist

- Functional Integrity: The application executes register, login, and protected operations without errors.
- API Specifications: Swagger UI endpoints are fully operational and input/output models are validated via Pydantic.
- AI Integrations: Multi-turn chat conversations maintain context, and semantic document retrieval provides relevant context.
- DevOps Execution: The multi-container system successfully spins up locally via Docker Compose.
- Live Deployment: All project links are active, showing stable frontend builds on Vercel/Render and backend APIs on Render/AWS.

Presentation Structure & Faculty Questions

Teams presented their work in a structured 10–15 minute format, covering: Introduction, Problem Statement, Solution, Technology Stack, Architecture, Live Demo, Challenges, and Future Scope. Students were prepared to answer key faculty questions covering: FastAPI async concurrency, JWT signing, Docker layer caching, S3 file storage, vector embeddings, and RAG retrieval pipelines.

Team-wise Project Registry & Evaluation

The following registry documents the project titles, core technical solutions, and detailed evaluation comments for all 17 student teams. The evaluations incorporate examiner insights and panel assessments:

Team 1: Job Recommendation Platform

Team Members: Nithya M, Ashritha, Prathibha R

Technical Solution: A full-stack recommendation engine utilizing FastAPI, React 18, and Qdrant to perform semantic profile-to-job matching based on cosine similarity.

Evaluation Comments: Solid contribution. The team demonstrated strong cooperation in implementing the recommendation algorithm. The job-matching accuracy was highly commendable.

Team 2: Hospital Information Management System

Team Members: Shetty Udbhavi Manjunath, Rakshita C, Anjali S Prabhu

Technical Solution: An integrated React frontend and FastAPI backend with a LangChain-powered conversational bot to query department FAQs, staff directories, and booking schedules.

Evaluation Comments: Well executed. Excellent integration of hospital workflow queries. The team worked cohesively and delivered a very practical tool.

Team 3: Smart Job Recommendation Engine

Team Members: Pratham Kashyap, Shekar N, Harshith R, Sumeet S U

Technical Solution: An intelligent RAG search pipeline using FastEmbed to vectorize resumes and job listings, indexed in a Qdrant Cloud instance, with secure JWT-based access.

Evaluation Comments: Good performance. Strong technical implementation of embedding storage in Qdrant. The vector search query routines worked seamlessly.

Team 4: Job Recommendation Chatbot

Team Members: Harshith R, Akash S M, Mufeed, Supreeth

Technical Solution: An interactive AI chatbot built with LangChain conversational memory, querying an indexed jobs corpus in Qdrant, deployed on Render and Vercel.

Evaluation Comments: Active participation. The chatbot's session state management was handled very well, resulting in an interactive and responsive live demonstration.

Team 5: AI Mental Health Chatbot

Team Members: Mohammed Amin, Mueez Maqbool, Akshay H

Technical Solution: A secure chatbot platform featuring custom prompt guards and safety boundaries, designed to assist users with stress management and coping strategies.

Evaluation Comments: Excellent outcome. Mohammed Amin demonstrated outstanding leadership and technical skill. The chatbot's conversational safety guidelines met the highest expected standards.

Team 6: Student Notes Chatbot (NotesAI)

Team Members: Akash M S, Srajan M, Mohammed Najeeb, Rahul N S

Technical Solution: A Retrieval-Augmented Generation (RAG) system where notes are chunked, embedded, and queried in Qdrant, generating context-rich summaries via Groq LLM.

Evaluation Comments: Cohesive effort. Excellent document retrieval implementation. The team worked well together to build a highly functional academic helper tool.

Team 7: AI-Powered Career Prep Platform

Team Members: Neha, Amitha P, Ashrifa K, Yasha V

Technical Solution: A full-stack application providing automated resume screening, semantic skill-gap analysis, and mock career coaching using LangChain agents.

Evaluation Comments: Outstanding presentation. The team's presentation and explanation skills were exemplary, though a deeper understanding of underlying model parameters is encouraged for future steps.

Team 8: Smart Job Recommendation Engine

Team Members: Chaithanya H U, Bhagya, Sandhya B L, Keerthi Gondadgi

Technical Solution: A semantic search portal that extracts skills from user profiles, query-embeds them via FastEmbed, and ranks open roles by similarity in Qdrant.

Evaluation Comments: Satisfactory delivery. A well-structured database schema and clean integration of PostgreSQL operations with the React UI cards.

Team 9: AI Job Recommendation System

Team Members: K Deeksha, Ashmitha Shetty, Rayzil, Pooja P

Technical Solution: A career portal featuring an integrated chat-matching interface, JWT authorization, and multi-service deployment using Supabase PostgreSQL.

Evaluation Comments: Superior architecture. Demonstrated excellent understanding of database connectivity using Supabase PostgreSQL. Their architectural execution was superior and highly commendable.

Team 10: College FAQ Chatbot

Team Members: Mohit Patil, Theertha Edneer, P Abinandan, Niranjana Baburaj K

Technical Solution: FAQ chatbot utilizing a RAG pipeline on college guidelines and handbooks, hosted on Vercel (frontend) and Render (backend).

Evaluation Comments: Highly polished. The team developed an exceptionally polished College FAQ bot. The interface and user experience were highly appreciated by the panel.

Team 11: Student Notes Chatbot

Team Members: Gurunath Ready, Neminath Dubale, Sandesh Malapur, Varun N Yaligar

Technical Solution: An interactive notes repository with an embedded AI research assistant that answers queries directly from uploaded PDFs and notes.

Evaluation Comments: Steady progress. Great design of the document upload and processing pipeline. The backend services were responsive and stable.

Team 12: Hospital Information System (MedInfo)

Team Members: Sushan U, Navaneeth, Pawan, Adarsh

Technical Solution: A management console with role-based logins and a smart booking chat assistant for checking availability, doctors' lists, and services.

Evaluation Comments: Clear structure. Strong demonstration of role-based routing. The division of doctor and patient roles was clearly visible and secure.

Team 13: CarrierAI (Career Assistant)

Team Members: Akshay C P, Akshay H, Akhilesh A W, Akash Gopal Naik

Technical Solution: An AI-native career portal that provides automated CV checks, semantic career roadmap generation, and smart job recommendations.

Evaluation Comments: Outstanding performance. Akshay C P and the team delivered an outstanding performance on CarrierAI. The project showcased high-quality code and robust service routing.

Team 14: Student Notes ChatBot

Team Members: Gurunath Reddy, Neminath Ravindra, Sandesh Malapur, Varun Yaligar

Technical Solution: A localized document retrieval pipeline integrating FastAPI backend, Neon PostgreSQL metadata tracking, and Qdrant semantic storage.

Evaluation Comments: Functional implementation. Good implementation of indexing in PostgreSQL. The metadata association with vectors was handled cleanly.

Team 15: Research Paper Assistant

Team Members: Mokshith H C, Pavan, Shetty Aadarsh Ravi, Varun

Technical Solution: A research-focused RAG chat console that extracts and retrieves text chunks from scientific PDFs, summarizing findings using a Groq LLM.

Evaluation Comments: Good delivery. A strong presentation of text chunking and LLM summarization. The team successfully handled large scientific PDF files.

Team 16: Research Paper Assistant

Team Members: J Manasa Reddy, M Preety Devi, Dhruv Kumar, Santhosh V

Technical Solution: A multi-document assistant built with LangChain, FastEmbed, and Qdrant, deployed in Docker containers to Render and Vercel.

Evaluation Comments: Exceptional outcome. Preety Devi and Manasa were key contributors who drove the project to meet and exceed all expected outcomes. Exceptional delivery on the RAG pipeline.

Team 17: Resume Analyzer

Team Members: Prathithi M L, Hamsini T R, Lahari M, Siri B

Technical Solution: A resume scanner using FastAPI to parse PDFs, scoring skills, and suggesting improvements using custom system prompts.

Evaluation Comments: Clean presentation. Very clean UI layout and smooth PDF parser integration. The resume analysis metrics were clearly presented.

Conclusion & Recommendations

The Capstone Phase successfully demonstrated that the BE/B.Tech Artificial Intelligence & Machine Learning (AI/ML) students have developed practical engineering capabilities. All 17 teams completed their projects and established live deployment URLs, demonstrating competency in backend, frontend, database, AI modeling, and containerized deployment.

The evaluation conducted by the college's own faculty panel validated the engineering choices and architecture. The students are well-prepared to step into technical roles as Associate Software Engineers, Full Stack Developers, and AI Application Engineers.

Report Compiled and Submitted By:
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